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POPULATION STUDY ARTICLE

Combining developmental and sleep health measures for autism spectrum disorder screening: an ECHO study

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BACKGROUND: Sleep problems are reported for up to 80% of autistic individuals. We examined whether parsimonious sets of items derived from the Modified Checklist for Autism in Toddlers, Revised (M-CHAT-R) and the Brief Infant Sleep Questionnaire (BISQ) are superior to the standard M-CHAT-R in predicting subsequent autism spectrum disorder (ASD) diagnoses.
METHODS: Participants from 11 Environmental influences on Child Health Outcomes (ECHO) cohorts were included. We performed logistic LASSO regression models with 10-fold cross-validation to identify whether a combination of items derived from the M-CHAT-R and BISQ are superior to the standard M-CHAT-R in predicting ASD diagnoses.
RESULTS: The final sample comprised 1552 children. The standard M-CHAT-R had a sensitivity of 44% (95% CI: 34, 55), specificity of 92% (95% CI: 91, 94), and AUROC of 0.726 (95% CI: 0.663, 0.790). A higher proportion of children with ASD had difficulty falling asleep or resisted bedtime during infancy/toddlerhood. However, LASSO models revealed parental reports of sleep problems did not improve the accuracy of the M-CHAT-R in predicting ASD diagnosis.
CONCLUSION: While children with ASD had higher rates of sleep problems during infancy/toddlerhood, there was no improvement in ASD developmental screening through the incorporation of parent-report sleep metrics.

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IMPACT:

- Parental-reported sleep problems are common in autism spectrum disorder (ASD).
- We investigated whether the inclusion of parental-reports of infant/toddler sleep patterns enhanced the effectiveness of developmental screening for autism.
- We reported higher rates of difficulty falling asleep and resisting bedtime during infancy and toddlerhood among children later diagnosed with ASD; however, we did not find an improvement in ASD developmental screening through the incorporation of parent-report sleep metrics.
- In our sample, the standard M-CHAT-R had a sensitivity of 39% among children of mothers with government insurance compared with a sensitivity of 53% among children of mothers with employer-based insurance.

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INTRODUCTION

Autism spectrum disorder (ASD) is a heterogeneous neurodevelopmental disorder characterized by disruption in social interaction and communication and the presence of a restricted range of interests and repetitive behaviors.[1] Empirical evidence reliably demonstrates that early intervention (EI) services from birth to 36 months of age improve quality of life outcomes for autistic individuals.[2,3] However, the median age of ASD diagnosis in the United States (U.S.) is between 4 and 4.5 years of age. To address this issue, the American Academy of Pediatrics recommends developmental screening using standardized and validated tools, such as the Modified Checklist for Autism, Revised (M-CHAT-R), at various points during a child's early years.[4] The M-CHAT-R is a parent-report screening tool validated to assess risk for ASD in infants and toddlers from 16–30 months of age. The U.S. Council of Children with Disabilities recommends a sensitivity of 70–80% for developmental screening tools,[5] and the M-CHAT-R has been shown to meet this criterion. The M-CHAT-R demonstrated a sensitivity ranging from 83–85%, a specificity ranging from 95–99%, and a positive predictive value (PPV) of 46.5% in its norming sample.[6] Recent meta-analyses have reported a pooled sensitivity of 83%, a pooled specificity of 46%, and a pooled PPV ranging from 45–51% in community-based samples in which diagnostic evaluation methods included either clinical evaluation or gold-standard diagnostic tools, such as the Autism Diagnosis Observation Schedule (ADOS).[1] However, commonly used screening tools, such as the M-CHAT-R, may perform differently across populations with various characteristics, including race,[7,8] ethnicity,[9] caregiver response style,[7] educational attainment,[10] culture,[11] neighborhood deprivation,[7] and rurality.[12] These differences in tool performance can exacerbate existing disparities, particularly in terms of racial, ethnic, and socioeconomic inequalities, in timely diagnosis and access to EI services. Therefore, understanding how the M-CHAT-R performs across populations with different characteristics can inform the creation of future developmental screening tools. Further, despite the pressing public health relevance, there have been limited advancements in improving ASD developmental screening tools over the past 15 years.

Although not a part of the diagnostic criteria for ASD, sleep problems are reported for up to 80% of autistic individuals across the lifespan.[13] Based on parental reports, children with ASD tend to exhibit more bedtime resistance, bedtime anxiety, and night-time awakenings than children with non-ASD developmental delays and children without developmental concerns.[14,15] Rather than sleep difficulties preceding autistic traits, emerging research based on parental reports suggests that sleep problems may be part of early ASD symptomology.[14,16] For example, Krakowiak and colleagues found that 53% of children (2–5 years of age) with ASD had experienced sleep problems at least once compared with 46% of children with non-ASD developmental delays and 32% of children without developmental concerns.[15] Using a multimethod approach, other researchers found that preschool children with autism and children with non-ASD developmental delays showed more parent-reported sleep problems than children without developmental concerns.[17]

Sleep problems are also prevalent among groups at increased risk for ASD. In the U.S., approximately 1 of every 10 infants is born preterm (before 37 weeks' gestation).[18,19] A meta-analysis revealed that the overall prevalence rate for ASD among infants born preterm was 7% (95% CI: 4%, 9%).[20] A wealth of scientific reports have highlighted poor sleep health and high rates of sleep impairments among children born preterm.[21,22] Premature birth alters the physiological maturation of central nervous system processes underlying the consolidation of pathways responsible for the development of sleep and sleep-wake rhythms.[23] These processes are foundational for sleep ontogenesis, which emerges in the second trimester and is consolidated in the third trimester of pregnancy.[24] Premature infants demonstrate immature sleep patterns, such as reduced sleep duration, lower arousal threshold, advanced sleep phase, altered electroencephalographic spectral values, and increased risk for sleep breathing disorders.[25] Sleep abnormalities among preterm infants have been shown to persist throughout childhood and through emerging adulthood. For example, a recent review that examined the relationship between prematurity and sleep from 5–18 years of age found that, in comparison to full-term infants, premature infants had earlier bedtimes and lower sleep quality as evidenced by more nocturnal awakenings and more non-rapid eye movement.[22] Although prior research has not examined the association of sleep parameters among preterm infants with ASD outcomes, one study reported that increased stage 2 sleep or decreased slow wave sleep or both were associated with an increase in total behavioral/emotional difficulties, hyperactivity-inattention, and peer problems among school age children born preterm.[26]

Together, this evidence suggests that the inclusion of parental reports of infant and toddler sleep concerns could improve early developmental screening for ASD. However, recent research has yet to integrate questionnaires regarding sleep problems into ASD screening assessments, such as the M-CHAT-R, despite evidence that sleep problems may co-occur with early autism symptomology in young children. Given the importance of brevity in universal screening for ASD, it is imperative that the administration of screening questions does not add significantly to family and/or provider burden. Therefore, the objective of this analysis was to identify whether parsimonious sets of items derived from the M-CHAT-R combined with a subset of items from the Brief Infant Sleep Questionnaire (BISQ) are superior to the standard M-CHAT-R in predicting subsequent ASD diagnoses in a large, diverse national U.S. cohort of children.

Our secondary analyses examined how the standard M-CHAT-R and parsimonious sets of items derived from the M-CHAT-R combined with a subset of items from the BISQ function across populations with differing characteristics, including child sex, race, ethnicity, and preterm birth status. We hypothesize that including parental reports of infant and toddler sleep concerns with the M-CHAT-R will improve accuracy in predicting subsequent ASD diagnosis. Based on prior research within disadvantaged communities,[7–9] we also hypothesized that both the M-CHAT-R and the combination of selected items from the BISQ along with M-CHAT-R items would have lower accuracy among racial- or ethnic-minority children and among children of mothers with government insurance.

METHODS

Environmental influences on Child Health Outcomes (ECHO) Cohort

The ECHO Cohort is a National Institutes of Health (NIH)-funded research consortium in the U.S.[27]. ECHO Cycle 1 included 69 pediatrics cohorts. ECHO aims to investigate the effects of early-life environmental exposures on child health and development by focusing on five key areas of child health: prenatal, perinatal, and postnatal health; obesity; respiratory conditions, including asthma; neurodevelopment; and positive health/well-being.[28,29] Existing data collected under cohort-specific protocols prior to the ECHO program and data collected using the ECHO-wide cohort data collection protocol[30] (https://echochildren.org/echo-program-protocol/) were harmonized by the ECHO Data Analysis Center (DAC). Cohort-specific and/or central ECHO institutional review boards approved the protocols, and participants provided informed consent.

Participants

This analysis included all ECHO participants who had non-missing data on all 20 items in the M-CHAT-R before 36 months of age, BISQ data before 36 months, and ASD diagnosis data ($N = 1552$). Additionally, since our predictors were based on maternal reports and since several of the ECHO cohorts included in this analysis had singleton pregnancy as an inclusion

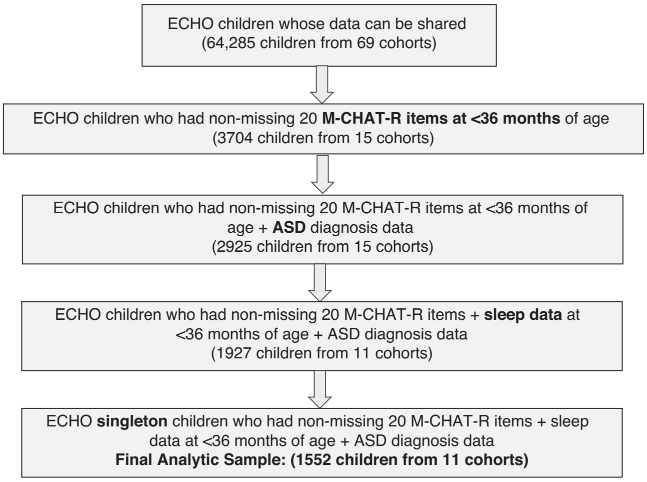

Fig. 1 Flow chart of participant selection.

criterion, we restricted participants to singletons. If a participant had multiple M-CHAT-R assessments, the last M-CHAT-R record before 36 months was selected for analysis. If multiple records of sleep data were available for a child, we chose the record that was the closest to the M-CHAT-R assessment completed prior to 36 months. A flow chart of our study sample selection is shown in Fig. 1.

ASD outcome

Our outcome variable was a binary indicator of whether a child has ever been diagnosed with ASD. This variable was harmonized by the ECHO DAC using data from the ECHO data collection forms and related extant cohort-specific data. For the majority of the sample (94%), ASD diagnosis data were collected from the child's medical history forms based on the question, "Has a doctor or other health care provider ever told you that the child has or had ASD?" For a smaller subset of participants from high-risk preterm birth cohorts (6% of the sample), ASD diagnosis was determined based on whether the child met ASD criteria on both the Autism Diagnostic Interview, Revised (ADI-R) and the Autism Diagnostic Observation Schedule 2nd edition (ADOS-2) under the Diagnostic and Statistical Manual, 5th Edition (DSM-V).

Predictors of ASD

The M-CHAT-R is a parent-report ASD screening tool validated for infants and toddlers aged 16–30 months. For most items on the M-CHAT-R, "yes" is a typical response, and "no" is an at-risk response. However, items 2, 5, and 12 are reverse scored, meaning that "no" is a typical response and "yes" is an at-risk response. When used clinically, M-CHAT-R items are summed, where a total score of 0–2 indicates "low risk," a total score of 3–7 indicates "moderate risk," and a total score of 8–20 indicates "high risk." For this analysis, instead of using the summed M-CHAT-R score, we utilized item-level M-CHAT-R questions (i.e., each question on the M-CHAT-R questionnaire; 20 items) (Fig. 2) and item-level BISQ questions (4 items) as the primary predictors of ASD. The BISQ is a parent-report measure of infant and toddler sleep problems validated for use from birth to 36 months of age. We selected 4 BISQ items based on data availability across cohorts. The BISQ items included in analysis assess (1) nighttime awakenings, defined as how often the child woke up more than one time during the night (never or sometimes); (2) difficultly falling asleep, defined as how often the child had difficulty or trouble falling asleep (never or almost never, sometimes, almost always or always); (3) resistance to bedtime, defined as how often the child resisted or intentionally put off bedtime (never or almost never, sometimes, almost always or always); and (4) excessive daytime sleepiness, defined as how often the child was excessively sleepy or tired during the daytime (never or almost never, sometimes, almost always or always). For sleep items with 3 categories, two dummy variables were created for "sometimes" and "almost always or always."

Characteristics of the study population

Maternal characteristics were collected through ECHO case report forms and medical record abstraction. They included age at delivery, race/ethnicity (Hispanic, Non-Hispanic White, Non-Hispanic Black, Non-Hispanic, or Other Race), education (less than high school, high school, some college, or college or above), marital status, maternal insurance status (private, Medicaid or Medicare, other, or none), delivery mode (vaginal or cesarean), and parity (0, 1, 2+). Child characteristics included child sex, race/ethnicity, and gestational age (GA) at birth. All variables used in this analysis were harmonized by the ECHO DAC.

Statistical analysis

Descriptive statistics were used to describe the characteristics of the study population. The chi-squared test of independence was used to determine whether infant and toddler sleep problems were increased among children who later received an ASD diagnosis. Polychoric correlation was used to measure the correlation between each M-CHAT-R and BISQ item.

We conducted a two-step analysis. The first step was to implement machine learning–based predictive modeling with the identified item-level features to determine a parsimonious set of predictor variables. Specifically, we performed two logistic least absolute shrinkage and selection operator (LASSO) regression models to predict binary ASD outcomes using cv.glmnet in R with 10-fold cross-validation.[31,32] LASSO Model 1 included 20 M-CHAT-R items as predictors, and Model 2 included the same 20 M-CHAT-R items from Model 1 with the addition of 4 BISQ items (i.e., nighttime awakenings, difficulty falling asleep, resistance to bedtime, excessive daytime sleepiness). To determine the most parsimonious model, we used the value of the regularization parameter lambda as plus one standard error of the minimum lambda when selecting the final model coefficients. The logistic LASSO regression was estimated in the overall study sample first and then stratified by participants' preterm birth status. Preterm birth was defined as a GA at birth < 37 weeks, and term birth was defined as a GA ≥ 37 weeks.

The second step was to compare the model performance of the two LASSO-selected models with the standard M-CHAT-R. That is, we ran the following 3 logistic models for ASD with predictors included: all 20 M-CHAT-R items (reference model), Model 1 (LASSO-selected M-CHAT-R items), and Model 2 (LASSO-selected M-CHAT-R and sleep items). The model performance was measured by the area under the receiver operating characteristic curve (AUROC), area under the precision-recall curve (AUPRC), sensitivity (true positive rate or recall), specificity (true negative rate), and positive predictive value (PPV). We evaluated the model performance measures for the overall study sample first and then applied the measures for subgroups stratified by child sex at birth and child's race and ethnicity. To calculate the prediction precision of the models, we adopted the same cutoff point of ≥ 3 of sum M-CHAT-R scores (moderate risk) as used in the formal M-CHAT-R scale and applied it to the items selected in each model for calculating model performance. The sensitivity and specificity of the standard M-CHAT-R and the two LASSO models were also calculated stratified by child-assigned sex at birth, child race and ethnicity, maternal insurance status, delivery mode, and GA category. Non-parametric Mann-Whitney tests were used to compare the AUROCs from the LASSO models to the standard M-CHAT-R.[33]

RESULTS

Participant characteristics

Our final sample comprised 1552 children born between 2002 and 2021 from 11 ECHO cohorts enrolled across 14 U.S. states (California, Colorado, Connecticut, Georgia, Hawaii, Illinois, Massachusetts, Michigan, Missouri, New York, North Carolina, Rhode Island, Tennessee, and Washington) and Puerto Rico (Fig. 1). Approximately half (51%) of the participants were enrolled into prospective pregnancy or childhood cohorts that recruited healthy pregnant persons or children, and approximately half (49%) of the participants were recruited into prospective childhood cohorts that recruited infants born preterm or low birth weight from neonatal intensive care units (NICUs). No ASD infant sibling cohorts were included in this analysis. For additional information regarding the inclusion and exclusion criteria for each cohort, please see the Supplemental Material. The mothers predominately identified as Non-Hispanic Black (40%), followed by Non-Hispanic White (35%), Hispanic (17%), and Non-Hispanic Other Race (7%). The majority of the mothers had some college education and above (75%), were married or living with a partner

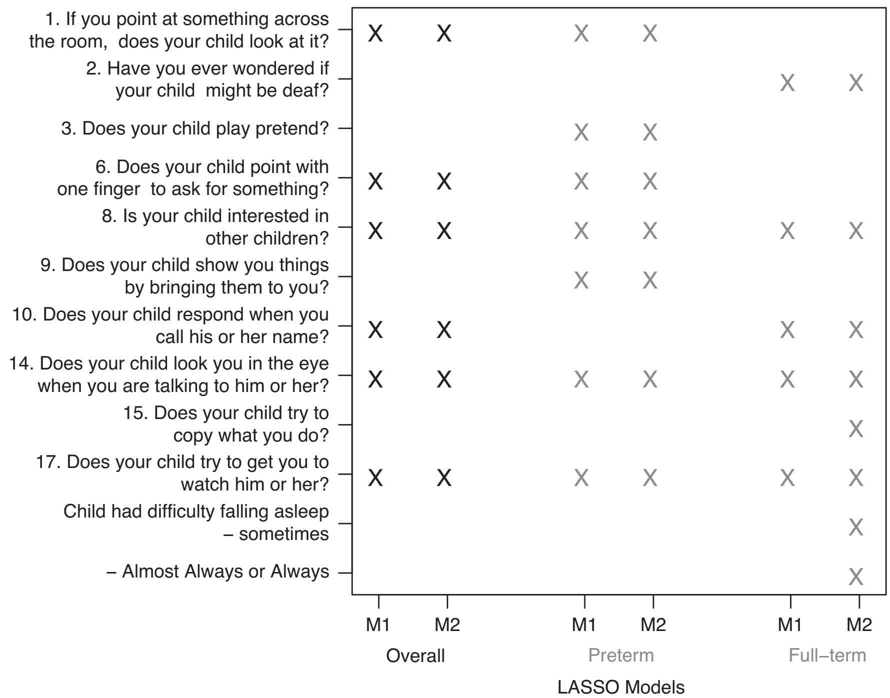

Fig. 2 Item-level features selected from the M-CHAT-R (M1, Model 1) and BISQ (M2, Model 2) among the overall sample, preterm sample, and full-term sample[a]. [a]Only items that were selected by at least one model are shown here. BISQ, Brief Infant Sleep Questionnaire; M-CHAT-R, Modified Checklist for Autism in Toddlers, Revised.

(63%), and had government insurance (52%) (Table 1). Slightly more than half (52%) of the children in this sample were born preterm (< 37 weeks' gestation) (Table 1). The M-CHAT-R assessments utilized in the present analysis were collected at approximately 26.3 ± 2.9 months of age, and the BISQ was administered at approximately 27.2 ± 3.9 months of age. Nearly 88% of the M-CHAT-R and BISQ assessments were completed within 1 month of each other (mean difference 0.96 ± 3.4 months; range: − 17 months to + 12 months, median = 0, IQR = 0, 0). ASD diagnostic outcomes (≥ age 4) were determined through medical record abstraction, parental report of a clinician's diagnosis, or a diagnostic evaluation. In the present sample, 6% of children had a lifetime diagnosis of ASD (Table 1). The exact age of ASD diagnosis was not available for 29% of individuals with an ASD diagnosis ($N = 29$). Among children for whom this information was available ($N = 63$), age at diagnosis ranged from 1 to 18 years of age (median = 4, IQR = 2, 8).

The proportion of children with an ASD diagnosis who had difficulty falling asleep ($p < .05$) or who resisted bedtime ($p < .05$) during infancy/toddlerhood were increased as compared to children without an ASD diagnosis (Table 2). However, we did not observe differences in increased nighttime awakenings or excessive daytime sleepiness between children who received an ASD diagnosis compared with children who did not receive an ASD diagnosis (p-values > 0.05) (Table 2).

Primary analyses

The standard M-CHAT-R had a sensitivity of 44% (95% CI: 34, 55), specificity of 92% (95% CI: 91, 94), PPV of 26% (95% CI: 19, 34), AUPRC of 0.295 (95% CI: 0.272, 0.318), and AUROC of 0.726 (95% CI: 0.663, 0.790) using an M-CHAT-R cutoff score of moderate risk (Table 3). LASSO Model 1, which included 20 M-CHAT-R items as predictors, identified 6 of the 20 M-CHAT-R item-level questions as predictors (Fig. 2). Items selected included information regarding joint attention, pointing, interest in other children, response to name, eye contact, and whether the child wants his or her parent to watch him or her. Items not selected included responses to sounds, imaginary play, the child showing his or her parent things, and copying behaviors (Fig. 2). Model 1 had a sensitivity of 18% (95% CI: 11, 28), specificity of 98% (95% CI: 98, 99), PPV of 42% (95% CI: 26, 59), AUPRC of 0.256 (95% CI: 0.234, 0.277), and AUROC of 0.700 (95% CI: 0.645, 0.755) (Table 3). Model 1 did not show a decreased AUROC compared with the standard 20-item model ($p > 0.05$) (Table 3). LASSO Model 2 (LASSO-selected M-CHAT-R [Model 1] and sleep items) did not identify any sleep items as adding predictive value for the ASD outcome (Fig. 3).

Stratified analyses

Models stratified by preterm status. Among children born preterm, the standard M-CHAT-R had a sensitivity of 45% (95% CI: 33, 57), specificity of 91% (95% CI: 89, 93), PPV of 32% (95% CI: 23, 42), AUPRC of 0.316 (95% CI: 0.284, 0.347), and AUROC of 0.718 (95% CI: 0.646, 0.790) using an M-CHAT-R cutoff score of moderate risk (Table 3). LASSO Model 1 identified 7 of the 20 M-CHAT-R item-level questions as predictors (Fig. 2). The items were the same as in overall Model 1, with the addition of imaginary play (Fig. 2). Model 1 had a sensitivity of 21% (95% CI: 12, 32), specificity of 97% (95% CI: 95, 98), PPV of 38% (95% CI: 23, 54), AUPRC of 0.279 (95% CI: 0.248, 0.31), and AUROC of 0.704 (95% CI: 0.641, 0.767) (Table 3). Among preterm children, Model 1 did not show a decreased AUROC compared with the standard M-CHAT-R ($p > .05$) (Table 3). Additionally, the LASSO models did not select any sleep items as predictive of ASD in the preterm subsample (Fig. 2).

Among children born at term, the standard M-CHAT-R had a sensitivity of 41% (95% CI: 18, 67), specificity of 94% (95% CI: 92, 95), PPV of 14% (95% CI: 6, 26), AUPRC of 0.502 (95% CI: 0.466, 0.539), and AUROC 0.813 (95% CI: 0.685, 0.941) using an M-CHAT-R cutoff score of moderate risk (Table 3). Among children born term, LASSO Model 1 identified 5 of the 20 M-CHAT-R item-level questions as predictors (Fig. 2). Items selected included responses to sounds, interest in other children, response to name, eye contact, and whether the child wants his or her parent to watch him or her (Fig. 2). Model 1 had a sensitivity of 24% (95% CI: 7, 50),

Table 1. Participant characteristics: demographic information, risk factors, and M-CHAT-R scores.

Sample characteristics	$N = 1552$
Maternal characteristics	
Age at delivery, *N* (%) with non-missing data	1517 (97.7%)
Mean (SD)	28 (6)
Median (IQR)	28 (23, 33)
Race and ethnicity, *N* (%) with non-missing data	1532 (98.7%)
Hispanic	264 (17%)
Non-hispanic white	541 (35%)
Non-hispanic black	617 (40%)
Non-hispanic other race	110 (7%)
Education, *N* (%) with non-missing data	1542 (99.4%)
<High school	122 (8%)
High school	265 (17%)
Some college and above	1155 (75%)
Marital status, *N* (%) with non-missing data	1484 (95.6%)
Married or living with a partner	934 (63%)
Insurance status, *N* (%) with non-missing data	1461 (94.1%)
Private	535 (37%)
Medicaid or medicare	766 (52%)
Other or none	160 (11%)
Delivery mode, *N* (%) with non-missing data	1467 (94.5%)
Vaginal	743 (51%)
Cesarean	724 (49%)
Parity, *N* (%) with non-missing data	917 (59.1%)
0	390 (43%)
1	288 (31%)
2+	239 (26%)
Child characteristics	
Sex, *N* (%) with non-missing data	1552 (100%)
Female	761 (49%)
Male	791 (51%)
Child race and ethnicity, *N* (%) with non-missing data	1544 (99.5%)
Hispanic	297 (19%)
Non-hispanic White	504 (33%)
Non-hispanic black	614 (40%)
Non-hispanic other race	129 (8%)
Gestational age at birth, *N* (%) with non-missing data	1552 (100%)
Mean (SD)	33 (7)
Median (IQR)	34 (26, 39)
Preterm birth (< 37 weeks' gestation)	820 (52.5)
Lifetime ASD diagnosis, *N* (%) with non-missing data	1552 (100%)
Yes	88 (6%)
M-CHAT-R scores for children with an ASD diagnosis, *N* (%) with non-missing data	88 (100%)
Low risk (0–2)	49 (56%)
Moderate risk (3–7)	25 (28%)
High risk (8–20)	14 (16%)
M-CHAT-R scores for children without an ASD diagnosis, *N* (%) with non-missing data	1464 (100%)
Low Risk (0–2)	1352 (92%)
Moderate risk (3–7)	90 (6%)
High risk (8 – 20)	22 (2%)

ASD autism spectrum disorder, *IQR* interquartile range, *M-CHAT-R* Modified Checklist for Autism in Toddlers, Revised, *SD* standard deviation.

specificity of 100% (95% CI: 99, 100), PPV of 100% (95% CI: 40, 100), AUPRC of 0.325 (95% CI: 0.291, 0.359), and AUROC of 0.797 (95% CI: 0.676, 0.918) (Table 3, Fig. 3). Model 1 did not show a decreased AUROC compared with the standard M-CHAT-R among children born at term ($p > .05$) (Table 3). Among children born at term, LASSO Model 2 additionally identified difficulty falling asleep as predictive. Model 2 showed an increased AUROC and specificity compared with the standard M-CHAT-R. However, the increases in the AUROC in Model 2 did not meet the threshold for statistical significance ($p > .05$).

Models stratified by child assigned sex at birth. The standard M-CHAT-R and LASSO-selected M-CHAT-R models showed similar model performance between children assigned as male or female at birth (Table 4). The LASSO M-CHAT-R models did not show a decreased AUROC compared with the standard M-CHAT-R among male ($p > 0.05$) or female ($p > 0.05$) children.

Models stratified by child race and ethnicity. The standard M-CHAT-R and LASSO-selected M-CHAT-R models showed similar model performance across child racial and ethnic groups (Table 4). We observed that the standard M-CHAT-R had the highest AUROC among Hispanic children (0.84; 95% CI: 0.723, 0.957) followed by Non-Hispanic Black children (0.812; 95% CI: 0.72, 0.904) and lastly Non-Hispanic White children (0.722; 95% CI: 0.626, 0.819). The LASSO M-CHAT-R models did not show a decreased AUROC compared with the standard M-CHAT-R among Non-Hispanic-White children ($p > .05$) but showed a decreased AUROC in Hispanic ($p = 0.034$) and Non-Hispanic-Black ($p = 0.028$) children.

Models stratified by maternal insurance status. The standard M-CHAT-R and LASSO-selected M-CHAT-R models showed a lower AUROC (0.735; 95% CI: 0.647, 0.823) among children of mothers with Medicare or Medicaid compared with children of mothers with private, employer-paid insurance (0.842; 95% CI: 0.753, 0.931) (Table 4). The LASSO M-CHAT-R models did not show a decreased AUROC compared with the standard M-CHAT-R among children born to mothers with private insurance ($p > 0.05$) but showed a decreased accuracy among those with Medicare or Medicaid ($p = 0.015$).

Models stratified by mode of delivery. The standard M-CHAT-R and LASSO-selected M-CHAT-R models revealed similar model performance across mode of delivery (Table 4). We observed that the standard M-CHAT-R had the highest AUROC among children born via vaginal birth (0.772; 95% CI: 0.694, 0.851). The LASSO M-CHAT-R models showed a decreased AUROC compared with the standard M-CHAT-R among children born via vaginal birth ($p < 0.05$) but did not show a decreased AUROC among children born via cesarean delivery ($p > 0.05$).

DISCUSSION

In the present study, parent reports of infant and toddler sleep problems did not improve the accuracy of the M-CHAT-R. Similar to prior research,[13–15] we observed that the proportion of children who had difficulty falling asleep or who resisted bedtime during infancy or toddlerhood was increased in children with ASD compared with children without an ASD diagnosis. However, there are several possible explanations as to why we did not observe an increased AUROC of the M-CHAT-R when considering parent reports of infant and toddler sleep parameters. It is possible that parent reports of infant and toddler sleep measures did not accurately characterize sleep. For example, there is some evidence that parents may overreport their child's sleep problems

Table 2. Proportion of infant or toddler sleep problems among children with and without ASD.

BISQ items	No ASD Diagnosis (*N* = 1464)	ASD Diagnosis (*N* = *88*)	*P*-value
How often the child woke up > 1x during the night			0.095
Never of almost never, *N* (%)	1068 (73%)	57 (65%)	
Sometimes, *N* (%)	396 (27%)	31 (35%)	
How often the child had difficulty or trouble falling asleep			0.01
Never or almost never, *N* (%)	1084 (74%)	54 (61%)	
Sometimes, *N* (%)	323 (22%)	26 (30%)	
Almost always or always, *N* (%)	57 (4%)	8 (9%)	
How often the child resisted or intentionally put off bedtime			0.025
Never or almost never, *N* (%)	935 (64%)	49 (56%)	
Sometimes, *N* (%)	436 (30%)	27 (31%)	
Almost always or always, *N* (%)	93 (6%)	12 (14%)	
How often the child was excessively sleepy or tired during the daytime			0.927
Never or almost never, N (%)	1279 (87%)	76 (86%)	
Sometimes, *N* (%)	165 (11%)	<15 (<17%)	
Almost always or always, *N* (%)	20 (1%)	<5 (<16%)	

ASD autism spectrum disorder, *BISQ* Brief Infant Sleep Questionnaire.

compared with more objective measures acquired via wearables, such as actigraphy, which measure physiological or activity parameters of sleep.[34] Previous literature has also reported possible bias effects, with parents of autistic children being more sensitive to disturbances during the night than parents of children without other developmental disorders, for a variety of reasons.[35] It is also possible that the limited available BISQ questions, which focused on nighttime awakenings, difficulty falling asleep, resistance to bedtime, and excessive daytime sleepiness, restrained our ability to comprehensively characterize infant and toddler sleep. Finally, it is also possible that, at the time of screening, the sleep problems reported in our sample were simply not significant enough to warrant an improvement in the sensitivity or specificity of the M-CHAT-R.

Although the inclusion of parent reports of infant and toddler sleep problems did not improve the accuracy of the M-CHAT-R, the LASSO M-CHAT-R model had an AUROC similar to that of the standard M-CHAT-R. The M-CHAT-R items identified across the LASSO M-CHAT-R model as being predictive of a subsequent ASD diagnosis included questions about joint attention, pointing, interest in other children, response to name, eye contact, and whether the child wants their parent to watch him or her. The M-CHAT-R items identified across the LASSO M-CHAT-R model as being predictive of a subsequent ASD diagnosis included questions about joint attention, pointing, interest in other children, response to name, eye contact, and whether the child wants their parent to watch them. Items not selected included responses to sounds, imaginary play, the child showing his or her parent things, and copying behaviors. There were some differences in LASSO M-CHAT-R items selected between children born term and children born preterm. Specifically, for children born preterm, the LASSO M-CHAT-R model selected each of the items described above with the addition of imaginary play. On the contrary, the LASSO M-CHAT-R model for children born term selected only 5 of the 20 M-CHAT-R items, which included responses to sounds, interest in other children, response to name, eye contact, and whether the child wants their parent to watch them. These findings suggest that developmental screening for ASD could be improved through either developing specific assessments for preterm versus term children, through tailoring questions based on gestational age category, or through incorporating gestational age at birth in scoring algorithms.

The U.S. Council of Children with Disabilities recommends a sensitivity of 70–80% for developmental screening tools[5] to ensure that at-risk children receive further objective evaluation or EI services if clinically indicated. The M-CHAT-R is one of the most widely used developmental screening tools to identify autism risk.[1,36] Therefore, understanding how the M-CHAT-R performs across population characteristics, such as racial and ethnic groups, GA categories, and socioeconomic factors, has the capacity to inform future developmental screening tools. In our sample, the standard M-CHAT-R had a sensitivity of only 44%, which is significantly lower than that recommended by the U.S. Council of Children with Disabilities. Our observed sensitivity value was also significantly lower than those reported in recently published meta-analyses, each of which reported a pooled sensitivity of approximately 82%. Several of the included articles utilized different sources for ASD diagnosis (such as ADOS or ADOS-2, clinical evaluation, the use of a secondary screener, and medical record review).[1,36] Although we observed a similar sensitivity and specificity by racial and ethnic minority group, GA category, and sex in our sample, the standard M-CHAT-R had a sensitivity of 39% among children of mothers with government insurance compared with a sensitivity of 53% among children of mothers with employer-based insurance.

These findings have important implications regarding the relationship between socioeconomic disparities and early identification, EI, and long-term outcomes for children at risk for ASD. Prior research has demonstrated socioeconomic disparities in screening for ASD. For example, children whose families do not have health insurance are significantly less likely to present for a well-child visit; thus, they miss early screening opportunities in this setting.[37] A study examining retrospective chart reviews of approximately 1000 children in the U.S. found that children who attended clinics in primarily low-income areas had more inconsistency in screening and referral practices than children living in higher-income areas.[38] In higher-income areas, children with developmental delays or perceived behavioral problems may be flagged earlier than in lower-income areas[39,40] due to the greater resources available in wealthier school districts and/or greater awareness by parents and teachers. On the other hand, a lack of financial resources can limit parental access to information, education, and access to quality health care, as well as access to screening tools.[41] However, fewer studies have evaluated

Table 3. Primary model and models stratified by gestational age category.

Model type	AUROC (95% CI)	AUPRC (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)	PPV (95% CI)
Primary models[a]:					
Logistic regression with 20 M-CHAT-R items	0.726 (0.663, 0.790)	0.295 (0.272, 0.318)	44% (34, 55)	92% (91, 94)	26% (19, 34)
LASSO-selected M-CHAT-R items (Model 1)	0.700 (0.645, 0.755)	0.256 (0.234, 0.277)	18% (11, 28)	98% (98, 99)	42% (26, 59)
Models for preterm children[b]:					
Logistic regression with 20 M-CHAT-R items	0.718 (0.646, 0.790)	0.316 (0.284, 0.347)	45% (33, 57)	91% (89, 93)	32% (23, 42)
LASSO-selected M-CHAT-R items (Model 1)	0.704 (0.641, 0.767)	0.279 (0.248, 0.31)	21% (12, 32)	97% (95, 98)	38% (23, 54)
Models for full term children[c]:					
Logistic regression with 20 M-CHAT-R items	0.813 (0.685, 0.941)	0.502 (0.466, 0.539)	41% (18, 67)	94% (92, 95)	14% (6, 26)
LASSO-selected M-CHAT-R items (Model 1)	0.797 (0.676, 0.918)	0.325 (0.291, 0.359)	24% (7, 50)	100% (99, 100)	100% (40, 100)
LASSO-selected M-CHAT-R and BISQ items (Model 2)	0.843 (0.723, 0.958)	0.374 (0.339, 0.409)	35% (14, 62)	99% (98, 100)	55% (23, 83)

[a]Total participants ($N = 1552$); participants with autism spectrum disorder ($N = 88$).
[b]Total participants ($N = 820$); participants with autism spectrum disorder ($N = 71$).
[c]Total participants ($N = 732$); participants with autism spectrum disorder ($N = 17$).
AUPRC area under the precision-recall curve, *AUROC* area under the receiver operating characteristic curve, *BISQ* Brief Infant Sleep Questionnaire, *CI* confidence interval, *LASSO* least absolute shrinkage and selection operator, *M-CHAT-R* Modified Checklist for Autism in Toddlers, Revised, *PPV* positive predictive value.

socioeconomic disparities in M-CHAT-R tool performance. Determining if the M-CHAT-R is similarly effective in screening for ASD risk across families from high and low socioeconomic status backgrounds is critical given prior reports of children from low-income families being more likely to receive an ASD diagnosis at a later age[42] and therefore less likely to receive EI services. These differences in tool performance can exacerbate existing socioeconomic inequalities in timely diagnosis and access to EI services. Therefore, future studies should attempt to develop objective autism screening tools or modify existing autism screening tools to function equitably across population characteristics.

Strengths and limitations

The strengths of these analyses include the leveraging of the ECHO Cohort to have a large sample size that included participants enrolled in 11 different cohorts as well as the geographic, racial and ethnic, and socioeconomic diversity of our sample. The majority of maternal participants self-identified as being part of a U.S. racial or ethnic minority group (65%) and had government insurance (52%). The participants were enrolled across 14 U.S. states and Puerto Rico, representing urban, rural, and underserved communities. Several limitations and potential sources of bias may have influenced our findings. Inclusion in the present analysis required non-missing data on the M-CHAT-R and BISQ and available information on ASD diagnosis; therefore, individuals with complete data may not accurately reflect ECHO at large. It is also possible that measurement error was introduced during the data harmonization process. In the case of multiple M-CHAT-R assessments, we selected the last M-CHAT-R record before 36 months for analysis since prior research indicates developmental screening for ASD improves as children age. It is possible that previously completing the M-CHAT-R may influence one's reporting when completing the M-CHAT-R for a second or third time. It is also possible that receiving an ASD diagnosis could influence parent-reporting on the M-CHAT-R.

LASSO regression was selected to enable the reduction in the number of variables in the models and due to its robustness to high variances. However, due to the modest number of children with ASD in our sample ($N = 88$), it is possible that our models had overfitting issues or collinearity. ASD diagnostic outcomes were determined through medical record abstraction, parental report, or diagnostic evaluations, which included the ADOS-2 and/or ADI-R; this may have resulted in over- or under-reporting of an ASD diagnosis. However, prior meta-analyses of community cohorts included manuscripts with a wide range of ASD diagnostic evaluation methods.[1,36] In our ECHO-wide harmonized data, the M-CHAT-R follow-up interview data were not available. In the absence of interview follow-up, the sensitivity values of the M-CHAT-R have ranged from 0.91–0.95 in community-based samples.[43] Future developmental screening studies should administer the M-CHAT-R with follow-up (M-CHAT-R/FU).[44] Finally, an additional limitation is that our sample did not have objective measures of infant and toddler sleep behaviors, which are more reliable than parent-report measures of infant and toddler sleep.[45,46]

CONCLUSION

In summary, while we observed that children with ASD had higher rates of difficulties falling asleep and resisting bedtime during infancy or toddlerhood, our study did not establish the improvement of early developmental screening for autism through the incorporation of parent-reported infant and toddler sleep metrics. However, the present study significantly contributes to the literature on structural disparities affecting early developmental screening for autism. We observed that maternal government insurance status was associated with a lower sensitivity on this commonly used autism screening tool, but importantly, maternal

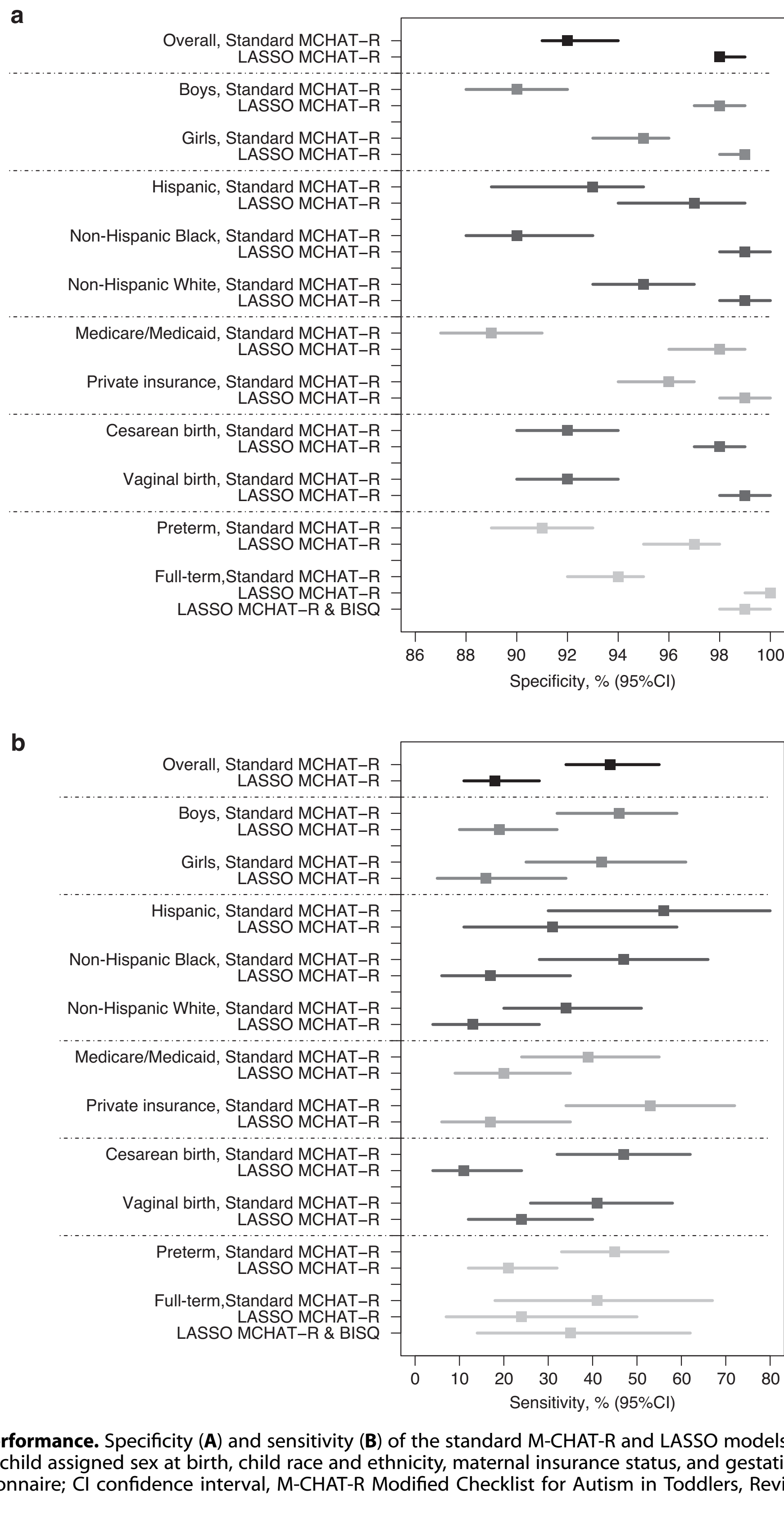

Fig. 3 Primary Model Performance. Specificity (**A**) and sensitivity (**B**) of the standard M-CHAT-R and LASSO models in the overall sample and the samples stratified by child assigned sex at birth, child race and ethnicity, maternal insurance status, and gestational age category[b]. [b]BISQ, Brief Infant Sleep Questionnaire; CI confidence interval, M-CHAT-R Modified Checklist for Autism in Toddlers, Revised.

race/ethnicity was not associated with screening outcomes. Future studies should consider examining novel methodologies, such as the development of objective screening tools, to improve developmental screening measures to function equitably across population characteristics and to move beyond parent-report measures. Finally, it is important to note that the M-CHAT-R items identified across several LASSO models are consistent with basic social communication, as opposed to motor, language, or sensory

Table 4. Models stratified by sociodemographic characteristics.

Model type	AUROC (95% CI)	AUPRC (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)	PPV (95% CI)
Stratified by Child Assigned Sex at Birth					
Male (Total, $N = 791$; ASD, $N = 57$)					
Logistic regression with 20 M-CHAT-R items	0.736 (0.66, 0.812)	0.344 (0.311,0.377)	46% (32, 59)	90% (88, 92)	26% (18, 36)
LASSO-selected M-CHAT-R items	0.698 (0.627, 0.768)	0.304 (0.272,0.336)	19% (10, 32)	98% (97, 99)	46% (26, 67)
Female (Total, $N = 761$; ASD, $N = 31$)					
Logistic regression with 20 M-CHAT-R items	0.767 (0.663, 0.871)	0.30 (0.268, 0.332)	42% (25, 61)	95% (93, 96)	25% (14, 39)
LASSO-selected M-CHAT-R items	0.712 (0.621, 0.803)	0.226 (0.197, 0.256)	16% (5, 34)	99% (98, 99)	36% (13, 65)
Stratified by Race and Ethnicity					
Hispanic children (Total, $N = 297$; ASD, $N = 16$)					
Logistic regression with 20 M-CHAT-R items	0.84 (0.723, 0.957)	0.548 (0.513, 0.582)	56% (30, 80)	93% (89, 95)	30% (15, 49)
LASSO-selected M-CHAT-R items	0.746 (0.609, 0.883)	0.343 (0.31, 0.376)	31% (11, 59)	97% (94, 99)	38% (14, 68)
Non-Hispanic Black children (Total, $N = 614$; ASD, $N = 30$)					
Logistic regression with 20 M-CHAT-R items	0.812 (0.72, 0.904)	0.346 (0.313, 0.379)	47% (28, 66)	90% (88, 93)	20% (11, 31)
LASSO-selected M-CHAT-R items	0.733 (0.639, 0.828)	0.266 (0.235, 0.297)	17% (6, 35)	99% (98, 100)	42% (15, 72)
Non-Hispanic White children (Total, $N = 504$; ASD, $N = 38$)					
Logistic regression with 20 M-CHAT-R items	0.722 (0.626, 0.819)	0.301 (0.269, 0.333)	34% (20, 51)	95% (93, 97)	38% (22, 56)
LASSO-selected M-CHAT-R items	0.679 (0.596, 0.762)	0.305 (0.273, 0.337)	13% (4, 28)	99% (98, 100)	56% (21, 86)
Stratified by Maternal Insurance Status					
Medicare/Medicaid (Total, $N = 766$; ASD, $N = 41$)					
Logistic regression with 20 M-CHAT-R items	0.735 (0.647, 0.823)	0.251 (0.229, 0.273)	39% (24, 55)	89% (87, 91)	17% (10, 26)
LASSO-selected M-CHAT-R items	0.653 (0.572, 0.734)	0.168 (0.149, 0.186)	20% (9, 35)	98% (96, 99)	33% (16, 55)
Private insurance (Total, $N = 535$; ASD, $N = 30$)					
Logistic regression with 20 M-CHAT-R items	0.842 (0.753, 0.931)	0.567 (0.542, 0.592)	53% (34, 72)	96% (94, 97)	43% (27, 61)
LASSO-selected M-CHAT-R items	0.787 (0.697, 0.878)	0.429 (0.404, 0.453)	17% (6, 35)	99% (98, 100)	50% (19, 81)
Stratified by Mode of Delivery					
Vaginal birth (Total, $N = 743$; ASD, $N = 41$)					
Logistic regression with 20 M-CHAT-R items	0.772 (0.694, 0.851)	0.368 (0.344, 0.392)	41% (26, 58)	92% (90,94)	23% (14, 35)
LASSO-selected M-CHAT-R items	0.694 (0.612, 0.776)	0.275 (0.252, 0.297)	24% (12, 40)	99% (98, 100)	56% (31, 78)
Cesarean birth (Total, $N = 724$; ASD, $N = 45$)					
Logistic regression with 20 M-CHAT-R items	0.749 (0.666, 0.832)	0.308 (0.285, 0.331)	47% (32, 62)	92% (90, 94)	29% (19, 41)
LASSO-selected M-CHAT-R items	0.704 (0.627, 0.781)	0.247 (0.225, 0.268)	11% (4, 24)	98% (97, 99)	28% (10, 53)

ASD autism spectrum disorder, *AUPRC* area under the precision-recall curve, *AUROC* area under the receiver operating characteristic curve, *BISQ* Brief Infant Sleep Questionnaire, *CI* confidence interval, *LASSO* least absolute shrinkage and selection operator, *M-CHAT-R* Modified Checklist for Autism in Toddlers Revised, *PPV* positive predictive value.

items. Providers should be particularly attentive to these items and behaviors in surveillance and follow-up.

DATA AVAILABILITY

Select de-identified data from the ECHO Program are available through NICHD's Data and Specimen Hub (DASH). Information on study data not available on DASH, such as some Indigenous datasets, can be found on the ECHO study DASH webpage.

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